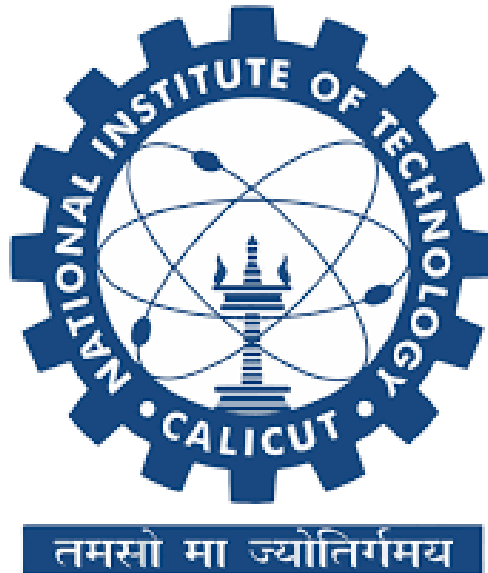


Microprocessors and Microcontrollers Course Project

AUTOMATIC NIGHT LAMP USING LDR AND PIC18F452



Submitted by

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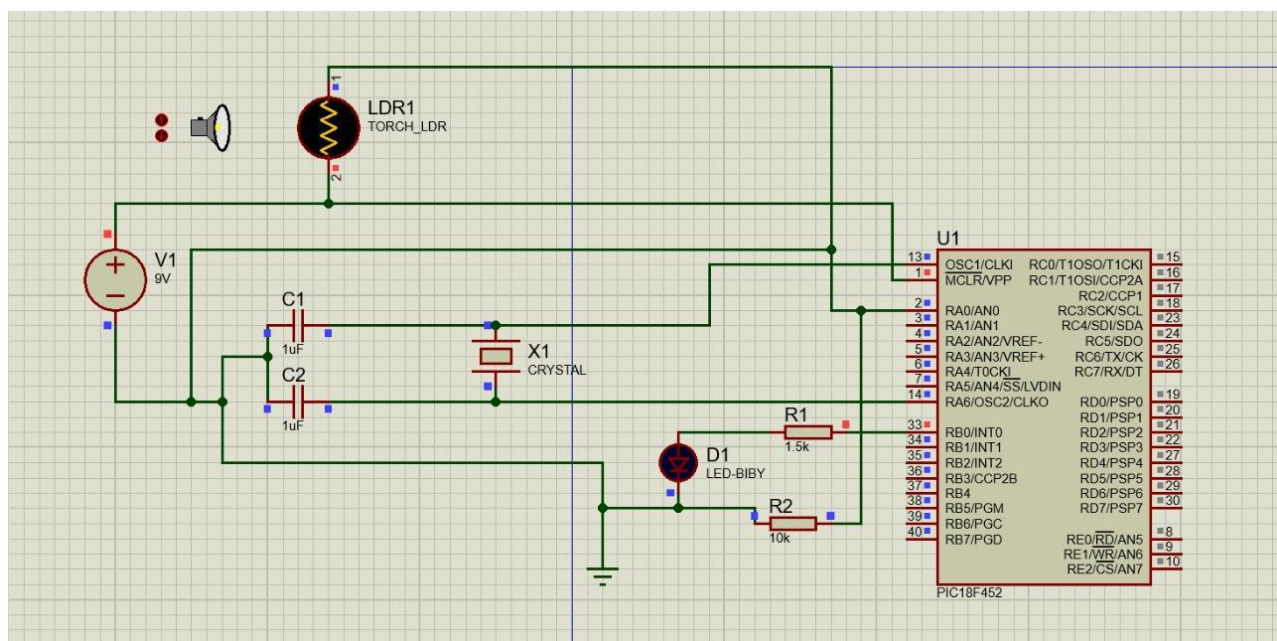
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OBJECTIVE OF THE PROJECT

The objective of this project is to design and implement an automatic night lamp using the PIC18F452 microcontroller and an LDR sensor. The system aims to automatically switch on an LED when the surrounding light intensity falls below a certain level, and turn it off when sufficient light is detected. This project demonstrates the use of the PIC18F452's analog-to-digital conversion feature for sensing light levels and making decisions accordingly. The main goal is to create a simple, low-cost, and energy-efficient lighting control system suitable for basic automation applications.

Sl. No.	Component	Specifications	Quantity
1	Power Supply	9 V DC	1
2	Voltage Regulator	7805 (5 V Output)	1
3	Microcontroller	PIC18F452	1
4	Crystal Oscillator	20 MHz	1
5	Capacitors	22 pF (for crystal)	2
6	LDR (Sensor)	Standard Light Sensor	1
7	Resistors	10 k Ω (for LDR divider) 1.5 k Ω (LED series)	1 1
8	LED (Streetlight)	Standard 5 mm LED	1
9	Breadboard	Full/half size	1
10	Connecting Wires	Male–Male Jumper Wires	5–10

CIRCUIT DIAGRAM



ASSEMBLY LANGUAGE CODE

```
#include <p18f452.inc>
```

```
; CONFIGURATION BITS
```

```
CONFIG OSC = HS    ; Use high-speed external crystal
```

```
CONFIG WDT = OFF   ; Disable Watchdog Timer
```

```
CONFIG LVP = OFF   ; Disable Low-Voltage Programming
```

```
; VARIABLE DECLARATION
```

```
CBLOCK 0x20        ; Start of general-purpose RAM
```

```
    COUNT1          ; Inner delay counter
```

```
    COUNT2          ; Middle delay counter
```

```
    COUNT3          ; Outer delay counter
```

```
ENDC
```

```
; RESET VECTOR
```

```
ORG 0x00           ; Program starts at reset address
```

```
    GOTO MAIN       ; Jump to main routine
```

```
; MAIN PROGRAM
```

```
MAIN:
```

```
    ; Configure PORTA as digital I/O
```

```
    MOVLW 0x06      ; Set RA0–RA3 as digital pins
```

```
    MOVWF ADCON1
```

```
    ; Set PORTA directions
```

```
    MOVLW b'00000001' ; RA0 as input, others as output
```

```
    MOVWF TRISA
```

```
    ; Set PORTB directions
```

```
    CLRF TRISB      ; All PORTB pins as output
```

```
    CLRF PORTB      ; Initialize PORTB to low
```

```
; MAIN LOOP
```

```
LOOP:
```

```
    MOVLW b'00000001' ; Prepare mask to read RA0
```

```
    XORWF PORTA, W    ; Compare RA0 with constant 1
```

```
    MOVWF PORTB       ; Output result on PORTB
```

```
    CALL DELAY        ; Call delay function
```

```
    GOTO LOOP         ; Repeat indefinitely
```

```
; DELAY SUBROUTINE
```

```
DELAY:
```

```
    MOVLW d'100'      ; Load outer counter
```

```
    MOVWF COUNT3
```

```
OUTERLOOP:
```

```
    MOVLW d'100'      ; Load middle counter
```

```
    MOVWF COUNT2
```

```
MIDDLELOOP:
```

```
    MOVLW d'50'       ; Load inner counter
```

```
    MOVWF COUNT1
```

```
INNERLOOP:
```

```
    NOP              ; Small time delay
```

```
    DECFSZ COUNT1, F ; Decrement inner counter
```

```
    GOTO INNERLOOP
```

```
DECFSZ COUNT2, F ; Decrement middle counter  
GOTO MIDDLELOOP  
DECFSZ COUNT3, F ; Decrement outer counter  
GOTO OUTERLOOP  
RETURN          ; Return to main loop
```

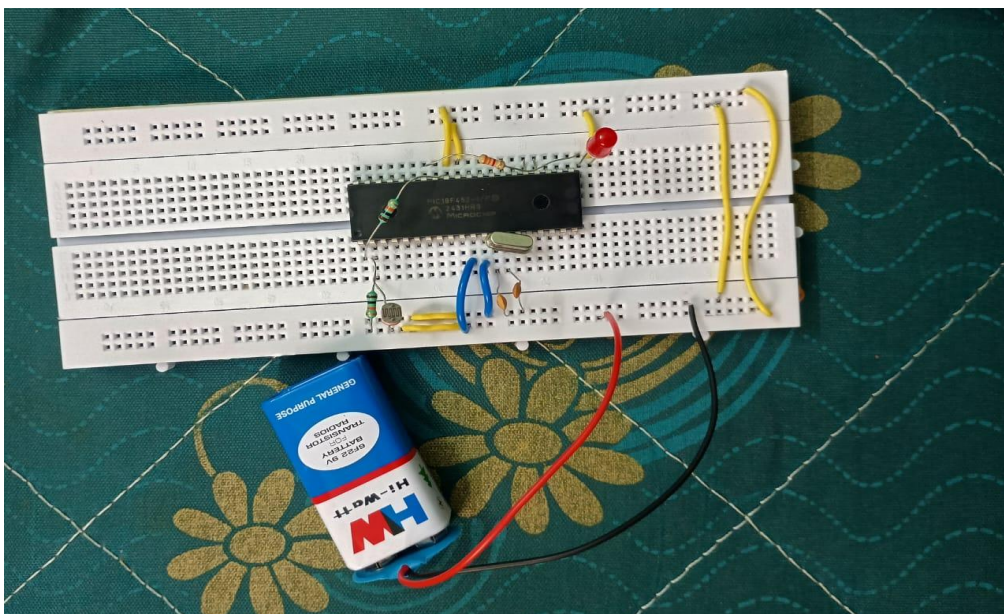
```
; END OF PROGRAM  
END
```

Circuit Connection

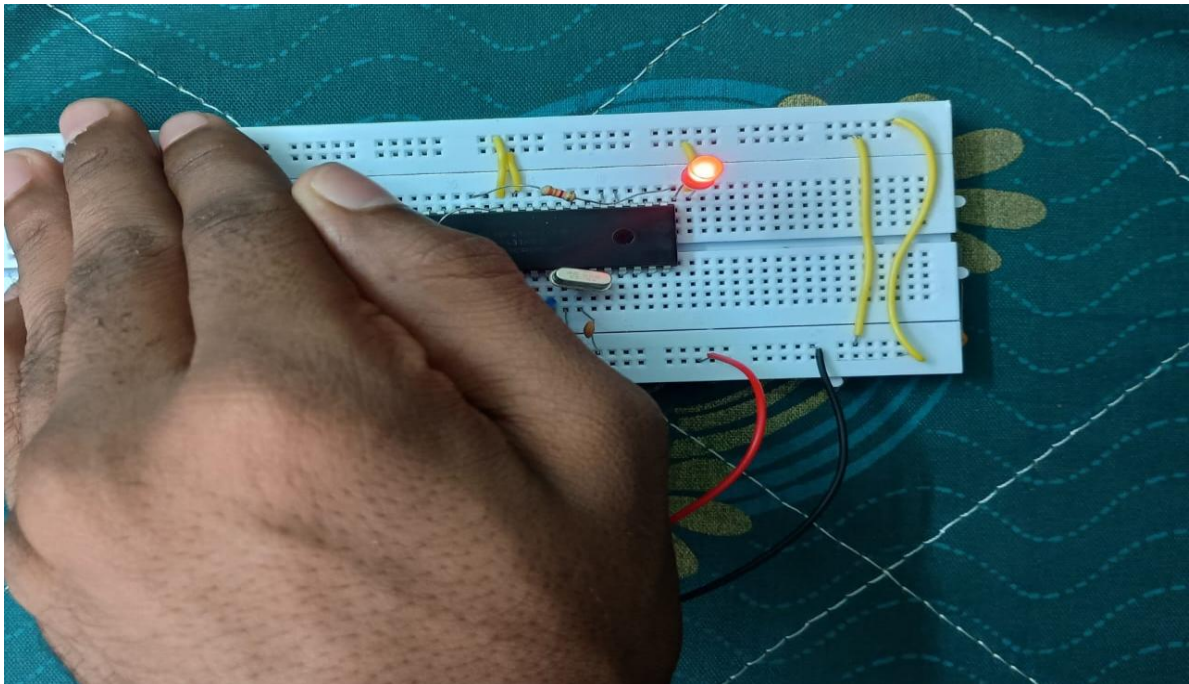
The circuit begins by forming a voltage divider using an LDR and a 10 k Ω resistor, with the junction point connected to the RA0 analog input pin of the PIC18F452. The LED used to represent the streetlight is connected to RB0 through an appropriate current-limiting resistor. A 20 MHz crystal oscillator is connected between the OSC1 and OSC2 pins of the microcontroller, accompanied by two capacitors to ground to ensure stable clock operation. The entire system is powered using a 9V DC supply regulated to 5V through a 7805 voltage regulator, which provides the required operating voltage for the PIC and associated components. All ground connections are properly linked to maintain stable and noise-free operation of the circuit.

OBSERVATIONS:

1. **The LED remains OFF** when the light intensity is above the threshold level, indicating sufficient ambient light and preventing unnecessary power usage.



2. The LED turns ON when the surrounding light intensity falls below the predefined threshold, indicating darkness and simulating automatic streetlight operation.



Conclusion:

The project successfully demonstrates automatic light control using an LDR and the PIC18F452 microcontroller. The system detects ambient light intensity and switches the LED accordingly—turning it ON in darkness and OFF in bright conditions. This simple yet effective setup highlights the use of ADC in microcontrollers for sensor-based automation and forms the basis for energy-efficient lighting applications.

REFERENCES:

- Microchip Technology Inc., *PIC18F452 Data Sheet – High-Performance Flash Microcontroller*, 2013.
- Mazidi, M.A., & McKinlay, R.D. (2008). *PIC Microcontroller and Embedded Systems: Using Assembly and C for PIC18*. Pearson Education.
- *Interfacing LDR Sensor with PIC Microcontroller*, EngineersGarage, <https://www.engineersgarage.com>